

Tushar Bhatt

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EDUCATION

KCC Institute of Technology and Management (AKTU) — New Delhi, India

Bachelor of Technology, Computer Science and Engineering

Aug 2024 – May 2028

BRMVB School — New Delhi, India

Higher Secondary Education, Non-Medical Science (PCM + CS)

Apr 2022 – Mar 2024

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, HTML/CSS, SQL

Frameworks: React, React Native (Expo), Node.js, Express.js, Flask, scikit-learn, MediaPipe, Pillow

Databases: MongoDB, MySQL

Tools: Git, GitHub, Docker, Google Colab, VS Code, Postman, EAS Build

Concepts: DSA, OOP, REST APIs, Machine Learning, Neural Networks, Web Automation

PROJECTS

ISL Bridge | React Native, Expo, Flask, MediaPipe, TensorFlow/Keras

2024 – Present

- Built a real-time Indian Sign Language (ISL) translator mobile app achieving 96–98% classification accuracy using a custom Dense Neural Network trained on 21-point MediaPipe hand landmark data.
- Designed a full-stack architecture with a React Native/Expo frontend and a Flask REST API backend (/health, /signs, /predict); handled multi-screen navigation (Splash, Onboarding, Permissions, Mode A/B).
- Engineered ML pipeline in Google Colab: data extraction, landmark normalization, model training, and weight serialization (weights.npz + label_map.json) to resolve cross-version Keras compatibility issues.
- Maintained two production GitHub repos (frontend + ML backend) with structured commit history; submitted to collegiate hackathon under Open Innovation track.

JARVIS — Local AI Assistant | Python, Ollama (Llama 3.2), MediaPipe, faster-whisper, edge-tts.

2024 – Present

- Built a fully offline AI assistant for Windows using Ollama (Llama3.2) for LLM inference, faster-whisper for speech recognition, and edge-tts for voice output — zero cloud dependency.
- Integrated MediaPipe hand tracking for gesture-based mode switching with multi-hand support, CRUD gesture management, and a custom gesture trainer module.
- Implemented pyautogui-based computer control pipeline enabling hands-free system interaction.

VoiceLegal | React, Node.js, Express, MongoDB Atlas, Web Speech API, GPT-4o

2024 – Present

- Built full-stack system with React/Tailwind frontend, Node/Express backend, MongoDB Atlas, and GitHub Models (GPT-4o) for LLM inference; scoped Google Maps API for nearest legal aid centres.

Architected an AI-powered Indian legal awareness assistant supporting Hindi and English voice input via Web Speech API, targeting underserved populations with limited legal literacy.

BotForge — Chatbot Factory Platform | HTML5, CSS3, JavaScript, Puter.js

2025 – Present

- Developed a futuristic chatbot-building platform featuring a cinematic landing page with interactive 3D mouse-follow effects, animated atom orbits, and dynamic particle backgrounds.
- Designed a premium dark-themed glassmorphism UI with smooth transitions, glowing visual effects, and responsive layouts to deliver a modern high-tech user experience.
- Built a step-by-step chatbot creation workflow enabling users to design, configure, train, and deploy custom AI chatbots through an intuitive interface.
- Integrated Puter.js to power backend interactions and streamline chatbot management functionalities directly within the browser environment.
- Engineered reusable animation systems and modular frontend components for scalable development and optimized rendering performance across devices.

Tunexa — Premium Media Downloader & Playlist Sync | React, Tailwind CSS, FastAPI, Python, FFmpeg, OAuth, yt-dlp

2026 – Present

- Developed a full-stack media downloader and playlist transfer platform supporting YouTube videos/playlists and Spotify tracks/playlists with a modern glassmorphism UI and smooth micro-animations.
- Engineered a FastAPI backend with yt-dlp and FFmpeg integration enabling 4K video downloads and 320kbps MP3 extraction with automatic audio-video stream merging.
- Implemented credentials-free Spotify metadata extraction and a playlist checklist system allowing selective multi-track downloads packaged into ZIP archives.
- Built secure OAuth-based Spotify → YouTube Music playlist transfer pipeline using ISRC matching and smart title-artist search heuristics for accurate song mapping.
- Designed self-cleaning backend utilities with in-memory token handling and automated temporary file cleanup to ensure secure and efficient local execution with zero mandatory cloud dependencies.

ACHIEVEMENTS

- LeetCode: Solved 100+ problems spanning arrays, trees, graphs, DP, and sliding window; consistent DSA practice for technical interviews.
- Hackathons: Competed under Team Alpha; participated in HackIndia 2026 (India's Biggest Web3 & AI Hackathon by C# Corner); consistently ranked in top 20% across multiple collegiate-level hackathons.
- GPA: Maintaining 8.5/10 in B.Tech CSE at AKTU while building production-level projects and running a live business.

CERTIFICATIONS

- be10x AI Tools & ChatGPT Workshop – Completion Certificate, be10x (Apr 2026)
- IEEE EDS MAIT – Certificate of Participation, Generative AI Webinar, Maharaja Agrasen Institute of Technology
- HackIndia 2026 – Certificate of Participation, India's Biggest Web3 & AI Hackathon, C# Corner (Team Alpha)
- HackerRank Voyager Blitz – Certificate of Participation, Voyager Blitz HackIndia Spark-4